

Chapter 1: Systems of Linear Equations

All Worked Examples - Step-by-Step Study PDF

Based on Ron Larson, Elementary Linear Algebra, 8th Edition, Chapter 1.

This PDF gathers the examples taught in the Chapter 1 tutoring session and expands the working so that every arithmetic step is easy to follow. Equations, matrices, substitutions, and final answers are placed in separate display blocks for clearer study.

Chapter sections covered: 1.1 Introduction to Systems of Linear Equations; 1.2 Gaussian Elimination and Gauss-Jordan Elimination; 1.3 Applications of Systems of Linear Equations.

Study method: read the problem, identify the type, follow each operation, then check the final answer by substitution.

Contents

1.1 - Introduction to Systems of Linear Equations

Example 1 - Linear and nonlinear equations

Example 2 - Parametric representation: $x_1 + 2x_2 = 4$

Example 3 - Parametric representation: $3x + 2y - z = 3$

Example 4 - Three possible solution types for two equations

Example 5 - Back-substitution in two variables

Example 6 - Back-substitution in three variables

Example 7 - Gaussian elimination in equation form

Example 8 - Inconsistent system

Applied Example - Balancing methane combustion

1.2 - Gaussian Elimination and Gauss-Jordan Elimination

Example 9 - Matrix size

Example 10 - Coefficient and augmented matrices

Example 11 - Elementary row operations

Example 12 - Gaussian elimination using matrices

Example 13 - Large system with four variables

Example 14 - Gauss-Jordan elimination

Example 15 - Free variable in reduced row-echelon form

Example 16 - Homogeneous system

1.3 - Applications of Systems of Linear Equations

Example 17 - Polynomial through three points

Example 18 - Polynomial through five points

Example 19 - Translating large x-values

Example 20 - Planetary period model

Example 21 - Network flow

Example 22 - Electrical network with three currents

Example 23 - Electrical network with six currents

1.1 Introduction to Systems of Linear Equations

A linear equation has variables only to the first power. A solution is a set of values that makes the equation true. A system is consistent when it has at least one solution and inconsistent when it has no solution.

General linear equation in n variables:

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b$$

Example 1 - Linear and Nonlinear Equations

Concept: Check whether variables appear only to the first power and are not multiplied together or placed inside nonlinear functions.

Classify each equation as linear or nonlinear.

- A. $3x + 2y = 7$
- B. $(1/2)x + y - \pi z = \sqrt{2}$
- C. $(\sin \pi)x_1 - 4x_2 = e^2$
- D. $xy + z = 2$
- E. $e^x - 2y = 4$
- F. $\sin(x_1) + 2x_2 - 3x_3 = 0$

Step 1: Check A

$$3x + 2y = 7$$

x and y are first power.

No xy , no x^2 , no $\sqrt{\text{variable}}$, no $\text{trig}(\text{variable})$.

Therefore A is linear.

Step 2: Check B

$$(1/2)x + y - \pi z = \sqrt{2}$$

$1/2$, π , and $\sqrt{2}$ are constants.

x , y , and z are first power.

Therefore B is linear.

Step 3: Check C

$$(\sin \pi)x_1 - 4x_2 = e^2$$

$\sin(\pi)$ is a number.

e^2 is a number.

x_1 and x_2 are first power.

Therefore C is linear.

Step 4: Check D

$$xy + z = 2$$

xy means x multiplied by y .

A product of variables is not allowed in a linear equation.

Therefore D is nonlinear.

Step 5: Check E

$$e^x - 2y = 4$$

x is in an exponential expression.

Therefore E is nonlinear.

Step 6: Check F

$$\sin(x_1) + 2x_2 - 3x_3 = 0$$

x_1 is inside a trigonometric function.

Therefore F is nonlinear.

Example 2 - Parametric Representation of a Solution Set

Concept: When one equation has two variables, choose one variable as free and write the other variable in terms of it.

Solve the linear equation.

$$x_1 + 2x_2 = 4$$

Step 1: Choose a free variable

Let $x_2 = t$, where t is any real number.

Step 2: Solve for x_1

$$x_1 + 2x_2 = 4$$

$$x_1 = 4 - 2x_2$$

Since $x_2 = t$:

$$x_1 = 4 - 2t$$

Step 3: Write the complete solution set

$$x_1 = 4 - 2t$$

$$x_2 = t$$

t is any real number.

Step 4: Check with $t = 1$

$$x_1 = 4 - 2(1)$$

$$x_1 = 2$$

$$x_2 = 1$$

Check:

$$2 + 2(1) = 4$$

$$2 + 2 = 4$$

$$4 = 4$$

Example 3 - Parametric Representation in Three Variables

Concept: With one equation in three variables, choose two variables as free variables.

Solve the linear equation.

$$3x + 2y - z = 3$$

Step 1: Choose free variables

Let $y = s$.

Let $z = t$.

s and t are any real numbers.

Step 2: Solve for x

$$3x + 2y - z = 3$$

$$3x = 3 - 2y + z$$

$$x = 1 - (2/3)y + (1/3)z$$

Step 3: Substitute $y = s$ and $z = t$

$$x = 1 - (2/3)s + (1/3)t$$

$$y = s$$

$$z = t$$

s and t are any real numbers.

Step 4: Check two particular solutions

If $s = 0$ and $t = 0$:

$$x = 1$$

$$y = 0$$

$$z = 0$$

Check:

$$3(1) + 2(0) - 0 = 3$$

$$3 = 3$$

If $s = 1$ and $t = 2$:

$$x = 1 - (2/3)(1) + (1/3)(2)$$

$$x = 1 - 2/3 + 2/3$$

$$x = 1$$

Check:

$$3(1) + 2(1) - 2 = 3$$

$$3 + 2 - 2 = 3$$

$$3 = 3$$

Example 4 - Systems of Two Equations in Two Variables

Concept: Two lines can intersect once, coincide, or be parallel. These give one solution, infinitely many solutions, or no solution.

Part A - Exactly one solution

Solve the system.

$$x + y = 3$$

$$x - y = -1$$

Step 1: Add the two equations

$$(x + y) + (x - y) = 3 + (-1)$$

$$x + x + y - y = 2$$

$$2x = 2$$

Step 2: Solve for x

$$2x = 2$$

$$x = 1$$

Step 3: Substitute into $x + y = 3$

$$1 + y = 3$$

$$y = 2$$

Final answer:

$$x = 1$$

$$y = 2$$

The system has exactly one solution.

Part B - Infinitely many solutions

Solve the system.

$$x + y = 3$$

$$2x + 2y = 6$$

Step 1: Compare the equations

$$2(x + y) = 2(3)$$

$$2x + 2y = 6$$

The second equation is the first equation multiplied by 2.

Step 2: Use one equation and a parameter

$$x + y = 3$$

Let $y = t$.

$$x + t = 3$$

$$x = 3 - t$$

Final answer:

$$x = 3 - t$$

$$y = t$$

t is any real number.

The system has infinitely many solutions.

Part C - No solution

Solve the system.

$$x + y = 3$$

$$x + y = 1$$

Step 1: Compare the equations

The left side is the same: $x + y$.
The right side is different: 3 and 1.

The same expression cannot equal 3 and 1 at the same time.

Final answer:

No solution.
The system is inconsistent.

Example 5 - Back-Substitution in Two Variables

Concept: Start from the last equation and work upward.

Solve the row-echelon system.

$$\begin{aligned}x - 2y &= 5 \\ y &= -2\end{aligned}$$

Step 1: Use the second equation

$$y = -2$$

Step 2: Substitute into the first equation

$$\begin{aligned}x - 2y &= 5 \\ x - 2(-2) &= 5 \\ x + 4 &= 5 \\ x &= 1\end{aligned}$$

Final answer:

$$\begin{aligned}x &= 1 \\ y &= -2\end{aligned}$$

Example 6 - Back-Substitution in Three Variables

Concept: Use the last equation first, then substitute upward.

Solve the row-echelon system.

$$\begin{aligned}x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ z &= 2\end{aligned}$$

Step 1: Use the third equation

$$z = 2$$

Step 2: Find y using the second equation

$$\begin{aligned}y + 3z &= 5 \\ y + 3(2) &= 5 \\ y + 6 &= 5 \\ y &= -1\end{aligned}$$

Step 3: Find x using the first equation

$$x - 2y + 3z = 9$$

$$x - 2(-1) + 3(2) = 9$$

$$x + 2 + 6 = 9$$

$$x + 8 = 9$$

$$x = 1$$

Final answer:

$$x = 1$$

$$y = -1$$

$$z = 2$$

Example 7 - Gaussian Elimination in Equation Form

Concept: Use equation operations to create row-echelon form, then back-substitute.

Solve the system.

$$x - 2y + 3z = 9$$

$$-x + 3y = -4$$

$$2x - 5y + 5z = 17$$

Step 1: Eliminate x from Equation 2

Equation 2 + Equation 1:

$$(-x + 3y) + (x - 2y + 3z) = -4 + 9$$

$$y + 3z = 5$$

Step 2: Eliminate x from Equation 3

Equation 3 + (-2)(Equation 1):

$$-2(x - 2y + 3z) = -2(9)$$

$$-2x + 4y - 6z = -18$$

Now add to Equation 3:

$$(2x - 5y + 5z) + (-2x + 4y - 6z) = 17 - 18$$

$$-y - z = -1$$

Step 3: Eliminate y from the new Equation 3

New Equation 3 + New Equation 2:

$$(-y - z) + (y + 3z) = -1 + 5$$

$$2z = 4$$

$$z = 2$$

Step 4: Back-substitute

$$z = 2$$

$$y + 3z = 5$$

$$y + 3(2) = 5$$

$$y = -1$$

$$x - 2y + 3z = 9$$

$$x - 2(-1) + 3(2) = 9$$

$$x + 8 = 9$$

$$x = 1$$

Final answer:

$$x = 1$$

$$y = -1$$

$$z = 2$$

Example 8 - An Inconsistent System

Concept: If elimination gives an impossible equation such as $0 = -2$, the system has no solution.

Solve the system.

$$x_1 - 3x_2 + x_3 = 1$$

$$2x_1 - x_2 - 2x_3 = 2$$

$$x_1 + 2x_2 - 3x_3 = -1$$

Step 1: Eliminate x_1 from Equation 2

Equation 2 + (-2) (Equation 1):

-2 (Equation 1):

$$-2x_1 + 6x_2 - 2x_3 = -2$$

Add to Equation 2:

$$(2x_1 - x_2 - 2x_3) + (-2x_1 + 6x_2 - 2x_3) = 2 - 2$$

$$5x_2 - 4x_3 = 0$$

Step 2: Eliminate x_1 from Equation 3

Equation 3 - Equation 1:

$$(x_1 + 2x_2 - 3x_3) - (x_1 - 3x_2 + x_3) = -1 - 1$$

$$5x_2 - 4x_3 = -2$$

Step 3: Compare the two new equations

$$5x_2 - 4x_3 = 0$$

$$5x_2 - 4x_3 = -2$$

Subtract the first from the second:

$$0 = -2$$

Final answer:

No solution.

The system is inconsistent.

Applied Example - Balancing Methane Combustion

Concept: Balancing a chemical equation can be written as a system of linear equations.

Balance the chemical equation.



Step 1: Write one equation for each atom type

Carbon atoms:

$$x_1 = x_3$$

Hydrogen atoms:

$$4x_1 = 2x_4$$

Oxygen atoms:

$$2x_2 = 2x_3 + x_4$$

Step 2: Choose a convenient value

Let $x_1 = 1$.

Then from $x_1 = x_3$:

$$x_3 = 1$$

Step 3: Find x_4 from hydrogen

$$4x_1 = 2x_4$$

$$4(1) = 2x_4$$

$$4 = 2x_4$$

$$x_4 = 2$$

Step 4: Find x_2 from oxygen

$$2x_2 = 2x_3 + x_4$$

$$2x_2 = 2(1) + 2$$

$$2x_2 = 4$$

$$x_2 = 2$$

Final balanced equation:



1.2 Gaussian Elimination and Gauss-Jordan Elimination

Section 1.2 uses matrices to solve systems without rewriting the variables at every step. The numbers are the same; the notation is shorter.

Augmented matrix idea:

[coefficients | constants]

Example 9 - Matrix Size

Concept: The size of a matrix is rows by columns: $m \times n$.

Part A

[2]

Rows = 1

Columns = 1

Size = 1×1

Part B

[0 0]

[0 0]

Rows = 2

Columns = 2

Size = 2×2

Part C

[e 2 -7]

[pi sqrt(2) 4]

Rows = 2

Columns = 3

Size = 2×3

Example 10 - Coefficient Matrix and Augmented Matrix

Concept: Write zero when a variable is missing.

Write the coefficient matrix and augmented matrix for the system.

$$\begin{aligned}x - 4y + 3z &= 5 \\ -x + 3y - z &= -3 \\ 2x - 4z &= 6\end{aligned}$$

Step 1: Align variables and insert a missing y coefficient

$$\begin{aligned}x - 4y + 3z &= 5 \\ -x + 3y - z &= -3 \\ 2x + 0y - 4z &= 6\end{aligned}$$

Step 2: Coefficient matrix

$$\begin{bmatrix} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{bmatrix}$$

Step 3: Augmented matrix

$$\begin{bmatrix} 1 & -4 & 3 & | & 5 \\ -1 & 3 & -1 & | & -3 \\ 2 & 0 & -4 & | & 6 \end{bmatrix}$$

Example 11 - Elementary Row Operations

Concept: Row operations produce row-equivalent matrices, meaning the solution set is unchanged.

Part A - Interchange two rows

Original matrix:

$$\begin{bmatrix} 0 & 1 & 3 & 4 \\ -1 & 2 & 0 & 3 \\ 2 & -3 & 4 & 1 \end{bmatrix}$$

Operation:

$R_1 \leftrightarrow R_2$

New matrix:

$$\begin{bmatrix} -1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 4 \\ 2 & -3 & 4 & 1 \end{bmatrix}$$

Part B - Multiply a row by a nonzero constant

Original matrix:

$$\begin{bmatrix} 2 & -4 & 6 & -2 \\ 1 & 3 & -3 & 0 \\ 5 & -2 & 1 & 2 \end{bmatrix}$$

Operation:

$(1/2)R_1 \rightarrow R_1$

New Row 1:

$(1/2)[2, -4, 6, -2] = [1, -2, 3, -1]$

New matrix:

$$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 1 & 3 & -3 & 0 \\ 5 & -2 & 1 & 2 \end{bmatrix}$$

Part C - Add a multiple of one row to another row

Original matrix:

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 2 & 1 & 5 & -2 \end{bmatrix}$$

Operation:

$$R3 + (-2)R1 \rightarrow R3$$

First calculate:

$$-2R1 = [-2, -4, 8, -6]$$

Then add to Row 3:

$$\begin{aligned} [2, 1, 5, -2] + [-2, -4, 8, -6] \\ = [0, -3, 13, -8] \end{aligned}$$

New matrix:

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 0 & -3 & 13 & -8 \end{bmatrix}$$

Example 12 - Gaussian Elimination Using Matrices

Concept: Use row operations to get row-echelon form, then back-substitute.

Solve the system.

$$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

Step 1: Write the augmented matrix

$$\begin{bmatrix} 1 & -2 & 3 & | & 9 \\ -1 & 3 & 0 & | & -4 \\ 2 & -5 & 5 & | & 17 \end{bmatrix}$$

Step 2: Eliminate x from Row 2

$$R2 + R1 \rightarrow R2$$

$$\begin{aligned} [-1, 3, 0, -4] + [1, -2, 3, 9] \\ = [0, 1, 3, 5] \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 3 & | & 9 \\ 0 & 1 & 3 & | & 5 \\ 2 & -5 & 5 & | & 17 \end{bmatrix}$$

Step 3: Eliminate x from Row 3

$$R3 + (-2)R1 \rightarrow R3$$

$$-2R1 = [-2, 4, -6, -18]$$

$$\begin{aligned} [2, -5, 5, 17] + [-2, 4, -6, -18] \\ = [0, -1, -1, -1] \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 3 & | & 9 \\ 0 & 1 & 3 & | & 5 \\ 0 & -1 & -1 & | & -1 \end{bmatrix}$$

Step 4: Eliminate y from Row 3

$$R_3 + R_2 \rightarrow R_3$$

$$\begin{bmatrix} 0 & -1 & -1 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 3 & 5 \end{bmatrix} \\ = \begin{bmatrix} 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

Step 5: Make the leading entry in Row 3 equal to 1

$$(1/2)R_3 \rightarrow R_3$$

$$\begin{bmatrix} 0 & 0 & 2 & 4 \end{bmatrix} \text{ becomes } \begin{bmatrix} 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Step 6: Back-substitute

$$z = 2$$

$$y + 3z = 5$$

$$y + 3(2) = 5$$

$$y = -1$$

$$x - 2y + 3z = 9$$

$$x - 2(-1) + 3(2) = 9$$

$$x = 1$$

Final answer:

$$x = 1$$

$$y = -1$$

$$z = 2$$

Example 13 - Gaussian Elimination with Four Variables

Concept: If a row begins with zero in the pivot column, swap rows to place a nonzero leading entry there.

Solve the system.

$$x_2 + x_3 - 2x_4 = -3$$

$$x_1 + 2x_2 - x_3 = 2$$

$$2x_1 + 4x_2 + x_3 - 3x_4 = -2$$

$$x_1 - 4x_2 - 7x_3 - x_4 = -19$$

Step 1: Write the augmented matrix

$$\begin{bmatrix} 0 & 1 & 1 & -2 & -3 \\ 1 & 2 & -1 & 0 & 2 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{bmatrix}$$

Step 2: Swap Row 1 and Row 2

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{bmatrix}$$

Step 3: Eliminate x_1 from Row 3

$$R3 + (-2)R1 \rightarrow R3$$

$$-2R1 = [-2, -4, 2, 0, -4]$$

$$[2, 4, 1, -3, -2] + [-2, -4, 2, 0, -4] \\ = [0, 0, 3, -3, -6]$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & | & 2 \\ 0 & 1 & 1 & -2 & | & -3 \\ 0 & 0 & 3 & -3 & | & -6 \\ 1 & -4 & -7 & -1 & | & -19 \end{bmatrix}$$

Step 4: Eliminate x_1 from Row 4

$$R4 + (-1)R1 \rightarrow R4$$

$$-R1 = [-1, -2, 1, 0, -2]$$

$$[1, -4, -7, -1, -19] + [-1, -2, 1, 0, -2] \\ = [0, -6, -6, -1, -21]$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & | & 2 \\ 0 & 1 & 1 & -2 & | & -3 \\ 0 & 0 & 3 & -3 & | & -6 \\ 0 & -6 & -6 & -1 & | & -21 \end{bmatrix}$$

Step 5: Eliminate x_2 from Row 4

$$R4 + 6R2 \rightarrow R4$$

$$6R2 = [0, 6, 6, -12, -18]$$

$$[0, -6, -6, -1, -21] + [0, 6, 6, -12, -18] \\ = [0, 0, 0, -13, -39]$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & | & 2 \\ 0 & 1 & 1 & -2 & | & -3 \\ 0 & 0 & 3 & -3 & | & -6 \\ 0 & 0 & 0 & -13 & | & -39 \end{bmatrix}$$

Step 6: Scale Row 3 and Row 4

$$(1/3)R3 \rightarrow R3$$

$$[0, 0, 3, -3, -6] \text{ becomes } [0, 0, 1, -1, -2]$$

$$(-1/13)R4 \rightarrow R4$$

$$[0, 0, 0, -13, -39] \text{ becomes } [0, 0, 0, 1, 3]$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & | & 2 \\ 0 & 1 & 1 & -2 & | & -3 \\ 0 & 0 & 1 & -1 & | & -2 \\ 0 & 0 & 0 & 1 & | & 3 \end{bmatrix}$$

Step 7: Back-substitute

$$x_4 = 3$$

$$x_3 - x_4 = -2$$

$$x_3 - 3 = -2$$

$$x_3 = 1$$

$$x_2 + x_3 - 2x_4 = -3$$

$$x_2 + 1 - 2(3) = -3$$

$$x_2 - 5 = -3$$

$$x_2 = 2$$

$$x_1 + 2x_2 - x_3 = 2$$

$$x_1 + 2(2) - 1 = 2$$

$$x_1 + 3 = 2$$

$$x_1 = -1$$

Final answer:

$$x_1 = -1$$

$$x_2 = 2$$

$$x_3 = 1$$

$$x_4 = 3$$

Example 14 - Gauss-Jordan Elimination

Concept: Continue beyond row-echelon form until zeros appear above and below every leading 1.

Solve the same three-variable system using Gauss-Jordan elimination.

$$x - 2y + 3z = 9$$

$$-x + 3y = -4$$

$$2x - 5y + 5z = 17$$

Step 1: Start from the row-echelon form

$$\begin{bmatrix} 1 & -2 & 3 & | & 9 \\ 0 & 1 & 3 & | & 5 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & | & 9 \\ 0 & 1 & 3 & | & 5 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & | & 9 \\ 0 & 1 & 3 & | & 5 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

Step 2: Clear the -2 above the leading 1 in column 2

$$R_1 + 2R_2 \rightarrow R_1$$

$$2R_2 = [0, 2, 6, 10]$$

$$[1, -2, 3, 9] + [0, 2, 6, 10]$$

$$= [1, 0, 9, 19]$$

$$\begin{bmatrix} 1 & 0 & 9 & | & 19 \\ 0 & 1 & 3 & | & 5 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 9 & | & 19 \\ 0 & 1 & 3 & | & 5 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 9 & | & 19 \\ 0 & 1 & 3 & | & 5 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

Step 3: Clear the 3 above the leading 1 in column 3

$$R_2 + (-3)R_3 \rightarrow R_2$$

$$-3R_3 = [0, 0, -3, -6]$$

$$[0, 1, 3, 5] + [0, 0, -3, -6]$$

$$= [0, 1, 0, -1]$$

$$\begin{bmatrix} 1 & 0 & 9 & | & 19 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 9 & | & 19 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 9 & | & 19 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

Step 4: Clear the 9 above the leading 1 in column 3

$$R1 + (-9)R3 \rightarrow R1$$

$$-9R3 = [0, 0, -9, -18]$$

$$\begin{aligned} [1, 0, 9, 19] + [0, 0, -9, -18] \\ = [1, 0, 0, 1] \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

Final answer read directly:

$$\begin{aligned} x &= 1 \\ y &= -1 \\ z &= 2 \end{aligned}$$

Example 15 - Free Variable in Reduced Row-Echelon Form

Concept: A nonleading variable becomes a parameter.

Solve the system.

$$\begin{aligned} 2x_1 + 4x_2 - 2x_3 &= 0 \\ 3x_1 + 5x_2 &= 1 \end{aligned}$$

Step 1: Write the augmented matrix

$$\begin{bmatrix} 2 & 4 & -2 & | & 0 \\ 3 & 5 & 0 & | & 1 \end{bmatrix}$$

Step 2: Use the reduced row-echelon form

$$\begin{bmatrix} 1 & 0 & 5 & | & 2 \\ 0 & 1 & -3 & | & -1 \end{bmatrix}$$

Step 3: Convert back to equations

$$\begin{aligned} x_1 + 5x_3 &= 2 \\ x_2 - 3x_3 &= -1 \end{aligned}$$

Step 4: Let the nonleading variable be free

$$\text{Let } x_3 = t.$$

Step 5: Solve for x_1 and x_2

$$\begin{aligned} x_1 + 5t &= 2 \\ x_1 &= 2 - 5t \end{aligned}$$

$$\begin{aligned} x_2 - 3t &= -1 \\ x_2 &= -1 + 3t \end{aligned}$$

Final answer:

$$\begin{aligned} x_1 &= 2 - 5t \\ x_2 &= -1 + 3t \\ x_3 &= t \end{aligned}$$

t is any real number.

Example 16 - Homogeneous System

Concept: A homogeneous system has all constant terms equal to zero. It always has the trivial solution.

Solve the homogeneous system.

$$\begin{aligned}x_1 - x_2 + 3x_3 &= 0 \\ 2x_1 + x_2 + 3x_3 &= 0\end{aligned}$$

Step 1: Write the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 2 & 1 & 3 & 0 \end{array} \right]$$

Step 2: Eliminate x_1 from Row 2

$$R_2 + (-2)R_1 \rightarrow R_2$$

$$-2R_1 = [-2, 2, -6, 0]$$

$$\begin{aligned}[2, 1, 3, 0] + [-2, 2, -6, 0] \\ = [0, 3, -3, 0]\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right]$$

Step 3: Scale Row 2

$$(1/3)R_2 \rightarrow R_2$$

$$[0, 3, -3, 0] \text{ becomes } [0, 1, -1, 0]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

Step 4: Clear the -1 above the leading 1

$$R_1 + R_2 \rightarrow R_1$$

$$\begin{aligned}[1, -1, 3, 0] + [0, 1, -1, 0] \\ = [1, 0, 2, 0]\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

Step 5: Use a parameter

$$\begin{aligned}x_1 + 2x_3 &= 0 \\ x_2 - x_3 &= 0\end{aligned}$$

$$\text{Let } x_3 = t.$$

$$\begin{aligned}x_1 + 2t &= 0 \\ x_1 &= -2t\end{aligned}$$

$$\begin{aligned}x_2 - t &= 0 \\ x_2 &= t\end{aligned}$$

Final answer:

$$\begin{aligned}x_1 &= -2t \\ x_2 &= t \\ x_3 &= t\end{aligned}$$

t is any real number.

When $t = 0$, the solution is $(0, 0, 0)$, the trivial solution.
When $t = 1$, the solution is $(-2, 1, 1)$, a nontrivial solution.

1.3 Applications of Systems of Linear Equations

Applications in this section use the same core idea: translate the real problem into equations, then solve the resulting system.

Example 17 - Polynomial Through Three Points

Concept: For n points, use a polynomial of degree $n - 1$.

Find a quadratic polynomial through the points.

$(1, 4), (2, 0), (3, 12)$

Step 1: Write the general quadratic polynomial

$$p(x) = a_0 + a_1x + a_2x^2$$

Step 2: Substitute each point

Point $(1, 4)$:

$$a_0 + a_1(1) + a_2(1)^2 = 4$$

$$a_0 + a_1 + a_2 = 4$$

Point $(2, 0)$:

$$a_0 + a_1(2) + a_2(2)^2 = 0$$

$$a_0 + 2a_1 + 4a_2 = 0$$

Point $(3, 12)$:

$$a_0 + a_1(3) + a_2(3)^2 = 12$$

$$a_0 + 3a_1 + 9a_2 = 12$$

Step 3: Solve the system

Equation 1:

$$a_0 + a_1 + a_2 = 4$$

Equation 2:

$$a_0 + 2a_1 + 4a_2 = 0$$

Equation 3:

$$a_0 + 3a_1 + 9a_2 = 12$$

Step 4: Subtract Equation 1 from Equation 2

$$(a_0 + 2a_1 + 4a_2) - (a_0 + a_1 + a_2) = 0 - 4$$

$$a_1 + 3a_2 = -4$$

Step 5: Subtract Equation 2 from Equation 3

$$(a_0 + 3a_1 + 9a_2) - (a_0 + 2a_1 + 4a_2) = 12 - 0$$

$$a_1 + 5a_2 = 12$$

Step 6: Eliminate a_1

$$(a_1 + 5a_2) - (a_1 + 3a_2) = 12 - (-4)$$

$$2a_2 = 16$$

$$a_2 = 8$$

Step 7: Find a_1

$$a_1 + 3a_2 = -4$$

$$a_1 + 3(8) = -4$$

$$a_1 + 24 = -4$$

$$a_1 = -28$$

Step 8: Find a_0

$$a_0 + a_1 + a_2 = 4$$

$$a_0 - 28 + 8 = 4$$

$$a_0 - 20 = 4$$

$$a_0 = 24$$

Final polynomial:

$$p(x) = 24 - 28x + 8x^2$$

or

$$p(x) = 8x^2 - 28x + 24$$

Example 18 - Polynomial Through Five Points

Concept: Five points require a fourth-degree polynomial.

Find a polynomial through the points.

$$(-2, 3), (-1, 5), (0, 1), (1, 4), (2, 10)$$

Step 1: Write the general fourth-degree polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

Step 2: Substitute each point

$$x = -2, y = 3:$$

$$a_0 - 2a_1 + 4a_2 - 8a_3 + 16a_4 = 3$$

$$x = -1, y = 5:$$

$$a_0 - a_1 + a_2 - a_3 + a_4 = 5$$

$$x = 0, y = 1:$$

$$a_0 = 1$$

$$x = 1, y = 4:$$

$$a_0 + a_1 + a_2 + a_3 + a_4 = 4$$

$$x = 2, y = 10:$$

$$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = 10$$

Step 3: Use $a_0 = 1$

$$\begin{aligned}-a_1 + a_2 - a_3 + a_4 &= 4 \\ a_1 + a_2 + a_3 + a_4 &= 3 \\ -2a_1 + 4a_2 - 8a_3 + 16a_4 &= 2 \\ 2a_1 + 4a_2 + 8a_3 + 16a_4 &= 9\end{aligned}$$

Step 4: Add the first two equations

$$\begin{aligned}(-a_1 + a_2 - a_3 + a_4) + (a_1 + a_2 + a_3 + a_4) &= 4 + 3 \\ 2a_2 + 2a_4 &= 7 \\ a_2 + a_4 &= 7/2\end{aligned}$$

Step 5: Subtract the first equation from the second

$$\begin{aligned}(a_1 + a_2 + a_3 + a_4) - (-a_1 + a_2 - a_3 + a_4) &= 3 - 4 \\ 2a_1 + 2a_3 &= -1 \\ a_1 + a_3 &= -1/2\end{aligned}$$

Step 6: Add the last two equations

$$\begin{aligned}(-2a_1 + 4a_2 - 8a_3 + 16a_4) + (2a_1 + 4a_2 + 8a_3 + 16a_4) &= 2 + 9 \\ 8a_2 + 32a_4 &= 11 \\ a_2 + 4a_4 &= 11/8\end{aligned}$$

Step 7: Find a_4 and a_2

$$\begin{aligned}a_2 + 4a_4 &= 11/8 \\ a_2 + a_4 &= 7/2 \\ \text{Subtract:} \\ 3a_4 &= 11/8 - 7/2 \\ 3a_4 &= 11/8 - 28/8 \\ 3a_4 &= -17/8 \\ a_4 &= -17/24 \\ \text{Now use } a_2 + a_4 &= 7/2: \\ a_2 - 17/24 &= 7/2 \\ a_2 &= 7/2 + 17/24 \\ a_2 &= 84/24 + 17/24 \\ a_2 &= 101/24\end{aligned}$$

Step 8: Subtract the third equation from the fourth

$$\begin{aligned}(2a_1 + 4a_2 + 8a_3 + 16a_4) - (-2a_1 + 4a_2 - 8a_3 + 16a_4) &= 9 - 2 \\ 4a_1 + 16a_3 &= 7 \\ a_1 + 4a_3 &= 7/4\end{aligned}$$

Step 9: Find a_3 and a_1

$$\begin{aligned}a_1 + 4a_3 &= 7/4 \\ a_1 + a_3 &= -1/2 \\ \text{Subtract:}\end{aligned}$$

$$3a_3 = 7/4 - (-1/2)$$

$$3a_3 = 7/4 + 2/4$$

$$3a_3 = 9/4$$

$$a_3 = 3/4$$

Now use $a_1 + a_3 = -1/2$:

$$a_1 + 3/4 = -1/2$$

$$a_1 = -1/2 - 3/4$$

$$a_1 = -2/4 - 3/4$$

$$a_1 = -5/4$$

Final polynomial:

$$p(x) = 1 - (5/4)x + (101/24)x^2 + (3/4)x^3 - (17/24)x^4$$

Example 19 - Translating Large x-Values Before Curve Fitting

Concept: If x-values are large, translate them to smaller numbers to make arithmetic easier.

Fit the data points.

(2011, 3), (2012, 5), (2013, 1), (2014, 4), (2015, 10)

Step 1: Use a translated variable

$$\text{Let } z = x - 2013$$

Step 2: Convert each x-value

$$x = 2011:$$

$$z = 2011 - 2013 = -2$$

$$x = 2012:$$

$$z = 2012 - 2013 = -1$$

$$x = 2013:$$

$$z = 2013 - 2013 = 0$$

$$x = 2014:$$

$$z = 2014 - 2013 = 1$$

$$x = 2015:$$

$$z = 2015 - 2013 = 2$$

Step 3: Use the polynomial from Example 18 in z

$$p(z) = 1 - (5/4)z + (101/24)z^2 + (3/4)z^3 - (17/24)z^4$$

Step 4: Substitute $z = x - 2013$

$$p(x) = 1$$

$$- (5/4)(x - 2013)$$

$$+ (101/24)(x - 2013)^2$$

$$+ (3/4)(x - 2013)^3$$

$$- (17/24)(x - 2013)^4$$

Example 20 - Planetary Period Model

Concept: Curve fitting can estimate values, but faraway points may not be predicted well.

Fit a quadratic to the three planets closest to the Sun and estimate the period of Mars.

Mercury: $x = 0.387$, $y = 0.241$
Venus: $x = 0.723$, $y = 0.615$
Earth: $x = 1.000$, $y = 1.000$
Mars: $x = 1.524$, $y = 1.881$

Step 1: Write the quadratic model

$$p(x) = a_0 + a_1x + a_2x^2$$

Step 2: Substitute Mercury, Venus, and Earth

$$a_0 + 0.387a_1 + (0.387)^2a_2 = 0.241$$

$$a_0 + 0.723a_1 + (0.723)^2a_2 = 0.615$$

$$a_0 + a_1 + a_2 = 1$$

Step 3: Compute the squares

$$(0.387)^2 = 0.149769$$

$$(0.723)^2 = 0.522729$$

So the system is:

$$a_0 + 0.387a_1 + 0.149769a_2 = 0.241$$

$$a_0 + 0.723a_1 + 0.522729a_2 = 0.615$$

$$a_0 + a_1 + a_2 = 1$$

Step 4: Use the approximate solution

$$a_0 \text{ approximately } -0.0634$$

$$a_1 \text{ approximately } 0.6119$$

$$a_2 \text{ approximately } 0.4515$$

Step 5: Write the model

$$p(x) = -0.0634 + 0.6119x + 0.4515x^2$$

Step 6: Estimate Mars using $x = 1.524$

$$p(1.524) = -0.0634 + 0.6119(1.524) + 0.4515(1.524)^2$$

$$0.6119(1.524) \text{ approximately } 0.9325$$

$$(1.524)^2 = 2.322576$$

$$0.4515(2.322576) \text{ approximately } 1.0487$$

$$p(1.524) \text{ approximately } -0.0634 + 0.9325 + 1.0487$$

$$p(1.524) \text{ approximately } 1.918$$

Comparison:

Estimated Mars period: 1.918 years

Actual table value: 1.881 years

The estimate is close, but it is not exact.

Example 21 - Network Flow

Concept: At every junction, total flow in equals total flow out.

Set up and solve the network system with five unknown flows.

$$\begin{aligned}x_1 + x_2 &= 20 \\x_3 - x_4 &= -20 \\x_2 + x_3 &= 20 \\x_1 - x_5 &= -10 \\-x_4 + x_5 &= -10\end{aligned}$$

Step 1: Let the free variable be x_5

$$\text{Let } x_5 = t.$$

Step 2: Use Equation 4

$$\begin{aligned}x_1 - x_5 &= -10 \\x_1 - t &= -10 \\x_1 &= t - 10\end{aligned}$$

Step 3: Use Equation 5

$$\begin{aligned}-x_4 + x_5 &= -10 \\-x_4 + t &= -10 \\-x_4 &= -10 - t \\x_4 &= t + 10\end{aligned}$$

Step 4: Use Equation 1

$$\begin{aligned}x_1 + x_2 &= 20 \\(t - 10) + x_2 &= 20 \\x_2 &= 30 - t\end{aligned}$$

Step 5: Use Equation 2

$$\begin{aligned}x_3 - x_4 &= -20 \\x_3 - (t + 10) &= -20 \\x_3 &= t - 10\end{aligned}$$

Step 6: Check Equation 3

$$\begin{aligned}x_2 + x_3 &= 20 \\(30 - t) + (t - 10) &= 20 \\20 &= 20\end{aligned}$$

Final answer:

$$\begin{aligned}x_1 &= t - 10 \\x_2 &= 30 - t \\x_3 &= t - 10 \\x_4 &= t + 10 \\x_5 &= t\end{aligned}$$

t is any real number.

If all flows must be nonnegative, then $10 \leq t \leq 30$.

Example 22 - Electrical Network with Three Currents

Concept: Use Kirchhoff law at junctions and voltage equations around loops.

Solve the current system.

$$\begin{aligned}I_1 - I_2 + I_3 &= 0 \\3I_1 + 2I_2 &= 7 \\2I_2 + 4I_3 &= 8\end{aligned}$$

Step 1: Solve the first equation for I_2

$$\begin{aligned}I_1 - I_2 + I_3 &= 0 \\I_1 + I_3 &= I_2 \\I_2 &= I_1 + I_3\end{aligned}$$

Step 2: Substitute into the second equation

$$\begin{aligned}3I_1 + 2I_2 &= 7 \\3I_1 + 2(I_1 + I_3) &= 7 \\3I_1 + 2I_1 + 2I_3 &= 7 \\5I_1 + 2I_3 &= 7\end{aligned}$$

Step 3: Substitute into the third equation

$$\begin{aligned}2I_2 + 4I_3 &= 8 \\2(I_1 + I_3) + 4I_3 &= 8 \\2I_1 + 2I_3 + 4I_3 &= 8 \\2I_1 + 6I_3 &= 8 \\I_1 + 3I_3 &= 4\end{aligned}$$

Step 4: Solve the two-equation system

$$\begin{aligned}5I_1 + 2I_3 &= 7 \\I_1 + 3I_3 &= 4\end{aligned}$$

From the second equation:

$$I_1 = 4 - 3I_3$$

Substitute into the first equation:

$$\begin{aligned}5(4 - 3I_3) + 2I_3 &= 7 \\20 - 15I_3 + 2I_3 &= 7 \\20 - 13I_3 &= 7 \\-13I_3 &= -13 \\I_3 &= 1\end{aligned}$$

Step 5: Find I_1 and I_2

$$\begin{aligned} I_1 &= 4 - 3I_3 \\ I_1 &= 4 - 3(1) \\ I_1 &= 1 \end{aligned}$$

$$\begin{aligned} I_2 &= I_1 + I_3 \\ I_2 &= 1 + 1 \\ I_2 &= 2 \end{aligned}$$

Final answer:

$$\begin{aligned} I_1 &= 1 \text{ amp} \\ I_2 &= 2 \text{ amps} \\ I_3 &= 1 \text{ amp} \end{aligned}$$

Example 23 - Electrical Network with Six Currents

Concept: Reduce a large circuit system by substitution, then solve the smaller system.

Solve the current system.

$$\begin{aligned} I_1 - I_2 + I_3 &= 0 \\ I_1 - I_2 + I_4 &= 0 \\ I_3 - I_5 + I_6 &= 0 \\ I_4 - I_5 + I_6 &= 0 \\ 2I_1 + 4I_2 &= 10 \\ 4I_2 + I_3 + 2I_4 + 2I_5 &= 17 \\ 2I_5 + 4I_6 &= 14 \end{aligned}$$

Step 1: Use the first equation

$$\begin{aligned} I_1 - I_2 + I_3 &= 0 \\ I_2 &= I_1 + I_3 \end{aligned}$$

Step 2: Use the second equation

$$\begin{aligned} I_1 - I_2 + I_4 &= 0 \\ \text{Substitute } I_2 &= I_1 + I_3: \\ I_1 - (I_1 + I_3) + I_4 &= 0 \\ -I_3 + I_4 &= 0 \\ I_4 &= I_3 \end{aligned}$$

Step 3: Use the third equation

$$\begin{aligned} I_3 - I_5 + I_6 &= 0 \\ I_5 &= I_3 + I_6 \end{aligned}$$

Step 4: Check the fourth equation

$$\begin{aligned} I_4 - I_5 + I_6 &= 0 \\ \text{Substitute } I_4 &= I_3 \text{ and } I_5 = I_3 + I_6: \\ I_3 - (I_3 + I_6) + I_6 &= 0 \\ 0 &= 0 \end{aligned}$$

This equation gives no new information.

Step 5: Use the fifth equation

$$2I_1 + 4I_2 = 10$$

Substitute $I_2 = I_1 + I_3$:

$$2I_1 + 4(I_1 + I_3) = 10$$

$$2I_1 + 4I_1 + 4I_3 = 10$$

$$6I_1 + 4I_3 = 10$$

$$3I_1 + 2I_3 = 5$$

Step 6: Use the sixth equation

$$4I_2 + I_3 + 2I_4 + 2I_5 = 17$$

Substitute:

$$I_2 = I_1 + I_3$$

$$I_4 = I_3$$

$$I_5 = I_3 + I_6$$

$$4(I_1 + I_3) + I_3 + 2(I_3) + 2(I_3 + I_6) = 17$$

$$4I_1 + 4I_3 + I_3 + 2I_3 + 2I_3 + 2I_6 = 17$$

$$4I_1 + 9I_3 + 2I_6 = 17$$

Step 7: Use the seventh equation

$$2I_5 + 4I_6 = 14$$

Substitute $I_5 = I_3 + I_6$:

$$2(I_3 + I_6) + 4I_6 = 14$$

$$2I_3 + 2I_6 + 4I_6 = 14$$

$$2I_3 + 6I_6 = 14$$

$$I_3 + 3I_6 = 7$$

Step 8: Solve the smaller system

$$3I_1 + 2I_3 = 5$$

$$4I_1 + 9I_3 + 2I_6 = 17$$

$$I_3 + 3I_6 = 7$$

Step 9: Express I_3 and I_1 in terms of I_6

From $I_3 + 3I_6 = 7$:

$$I_3 = 7 - 3I_6$$

Use $3I_1 + 2I_3 = 5$:

$$3I_1 + 2(7 - 3I_6) = 5$$

$$3I_1 + 14 - 6I_6 = 5$$

$$3I_1 = -9 + 6I_6$$

$$I_1 = -3 + 2I_6$$

Step 10: Find I_6

$$\text{Use } 4I_1 + 9I_3 + 2I_6 = 17:$$

$$4(-3 + 2I_6) + 9(7 - 3I_6) + 2I_6 = 17$$

$$-12 + 8I_6 + 63 - 27I_6 + 2I_6 = 17$$

$$51 - 17I_6 = 17$$

$$-17I_6 = -34$$

$$I_6 = 2$$

Step 11: Find all other currents

$$I_3 = 7 - 3I_6$$

$$I_3 = 7 - 3(2)$$

$$I_3 = 1$$

$$I_1 = -3 + 2I_6$$

$$I_1 = -3 + 2(2)$$

$$I_1 = 1$$

$$I_2 = I_1 + I_3$$

$$I_2 = 1 + 1$$

$$I_2 = 2$$

$$I_4 = I_3$$

$$I_4 = 1$$

$$I_5 = I_3 + I_6$$

$$I_5 = 1 + 2$$

$$I_5 = 3$$

Final answer:

$$I_1 = 1 \text{ amp}$$

$$I_2 = 2 \text{ amps}$$

$$I_3 = 1 \text{ amp}$$

$$I_4 = 1 \text{ amp}$$

$$I_5 = 3 \text{ amps}$$

$$I_6 = 2 \text{ amps}$$

Final Study Checklist

- Linear equation: variables appear only to the first power.
- Solution: values that make every equation true.
- Consistent system: at least one solution.
- Inconsistent system: no solution.
- One solution: leading variable for every variable.
- Infinitely many solutions: at least one free variable.
- No solution: a row or equation such as $0 = \text{nonzero}$.
- Gaussian elimination: row-echelon form, then back-substitution.
- Gauss-Jordan elimination: reduced row-echelon form, answer read directly.
- Homogeneous systems always have the trivial solution.
- Polynomial fitting: substitute points into the polynomial to form a system.
- Network flow: flow in equals flow out.
- Electrical networks: use Kirchhoff laws and solve the resulting system.